



Kennedy Baptist College
Semester One
Examination 2018
Question/Answer booklet

MATHEMATICS
METHODS UNIT 1

Section One:
Calculator-free

Student's Name: _____

SOLUTIONS

Teacher's Name: _____

Time allowed for this section

Reading time before commencing work: five minutes
Working time for paper: fifty minutes

Material required/recommended for this section

To be provided by the supervisor

This Question/Answer booklet
Formula Sheet

To be provided by the candidate

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener,
correction tape/fluid, erasers, ruler, highlighters

Special Items: nil

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised notes or other items of a non-personal nature in the examination room. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Section One: Calculator-free

35% (52 marks)

This section has **nine (9)** questions. Attempt **all** questions. Write your answers in the spaces provided.

Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.

- Planning: If you use the spare pages for planning, indicate this clearly at the top of the page.
- Continuing an answer: If you need to use the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number. Fill in the number of the question(s) that you are continuing to answer at the top of the page.

Working time: 50 minutes

Question 1 (6 marks)

Consider the function $g(x) = ax^2 + 11x + 6$ where a is a constant.

- (a) (i) Given that $g(4) = 98$, evaluate $g(-2)$. (2 marks)

$$\begin{aligned} 98 &= 16a + 44 + 6 \\ 98 &= 16a + 50 \\ 48 &= 16a \\ a &= 3 \quad \checkmark \end{aligned}$$

$$\begin{aligned} g(-2) &= 3(-2)^2 + 11(-2) + 6 \\ &= 12 - 22 + 6 \\ &= -4 \\ \therefore g(-2) &= -4 \quad \checkmark \end{aligned}$$

- (ii) State the equation of the axis of symmetry. (1 mark)

$$x = -\frac{11}{6} \quad \checkmark$$

- (iii) State the nature of the turning point. (1 mark)

$\uparrow \uparrow$ min. T. P. $\checkmark$

- (b) Determine all possible values of a given that $g(x)$ has no real solutions. (2 marks)

$$\begin{aligned} \Delta &< 0 \text{ (no real sol'n)} \\ b^2 - 4ac &< 0 \\ 121 - 24a &< 0 \quad \checkmark \\ -24a &< -121 \\ a &> \frac{121}{24} \quad \checkmark \end{aligned}$$

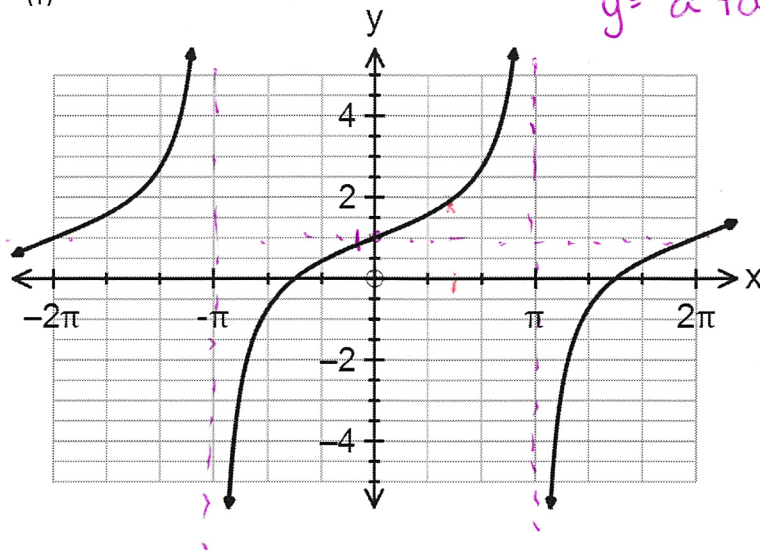
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Question 2 (6 marks)

(a) Find the equations of the relations below.

(i)



$$y = a \tan(bx) + c$$

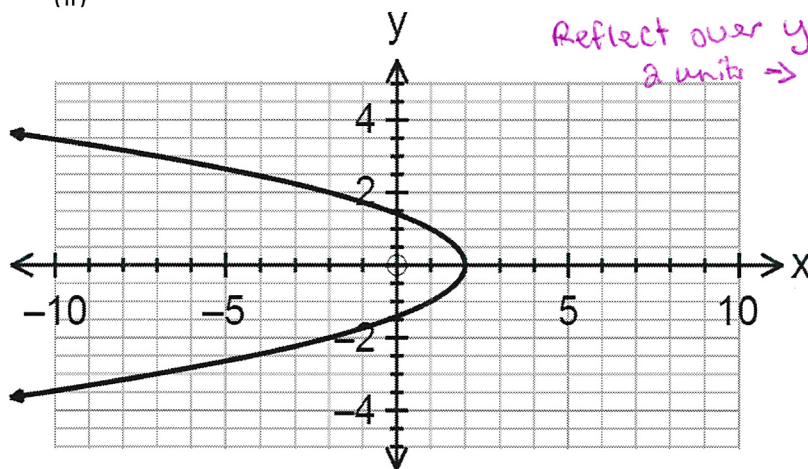
(2 marks)

$$P = \frac{\pi}{b}$$

$$b = \frac{\pi}{2\pi} = \frac{1}{2}$$

$$\therefore y = \tan \frac{1}{2}x + 1$$

(ii)



Reflect over y
2 units →

$$x = y^2$$



$$-x = y^2 \text{ Reflect}$$



$$-(x-2) = y^2 \text{ shift}$$

$$\text{or } \left. \begin{array}{l} y^2 = -x + 2 \\ x = -y^2 + 2 \end{array} \right\} \text{ Either}$$

(b) Explain which of the two relations above in (a) is a function and why.

(2 marks)

i) as there are no repeats of x coordinates



ii) has repeats ∴ is relation not

function

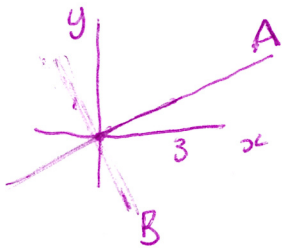
1/6

Question 3 (6 marks)

Line A and line B in the xy-plane intersect at 90° at the origin. Line A has a slope of $\frac{1}{3}$.

(a) Show that $P(2, -6)$ is a point on line B.

(2 marks)



$$m_A = \frac{1}{3} \therefore m_B = -3$$

$$\therefore y = -3x \text{ (B)} \checkmark$$

Test $(2, -6)$

$$-6 = -3(2)$$

$$-6 = -6 \text{ True } \therefore P \text{ is a point on B. } \checkmark$$

(b) Point $P(2, -6)$ is the midpoint of line segment CD which is parallel to Line A. Given that the x-value of C is -1 , find the coordinates of point D.

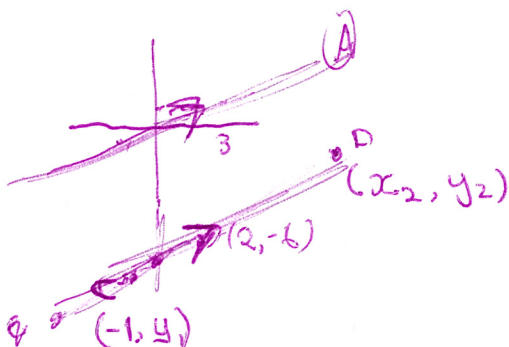
(4 marks)

Use midpoint $= \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$
 $(2, -6)$

$$2 = \frac{-1 + x_2}{2}$$

$$4 = -1 + x_2$$

$$x_2 = 5 \checkmark$$



$$m_{CD} = \frac{1}{3}$$

use $(2, -6)$

$$-6 = \frac{1}{3}(2) + c$$

$$c = -\frac{20}{3}$$

$$\therefore CD \Rightarrow y = \frac{1}{3}x - \frac{20}{3} \checkmark$$

$\therefore$ Point C when $x = -1$

$$y = \frac{1}{3}(-1) - \frac{20}{3}$$

$$= -\frac{21}{3} = -7 \quad C(-1, -7) \checkmark$$

See next page

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Question 4 (4 marks)

- (a) In a right triangle, one angle measures x° , where $\sin x = \frac{4}{5}$.

State the value of $\cos(90 - x^\circ)$.

(1 mark)



Identity
 $\cos(90 - x^\circ) = \frac{4}{5}$ ✓

- (b) Solve the equation $(\tan \theta + 1)(\sin^2 \theta - \sin \theta) = 0$ for θ , given that $-\pi \leq \theta \leq 2\pi$. (3 marks)

$$\tan \theta + 1 = 0 \quad \text{or} \quad \sin^2 \theta - \sin \theta = 0$$

$$\tan \theta = -1 \quad \text{or} \quad \sin \theta (\sin \theta - 1) = 0$$

$$\sin \theta = 0 \quad \text{or} \quad \sin \theta = 1$$



$$-\pi, 0, \pi, 2\pi$$



$$\frac{\pi}{2}$$



$$\theta = -\frac{\pi}{4}, \frac{3\pi}{4}, \frac{7\pi}{4}, \frac{\pi}{4}$$



4

Question 5 (3 marks)

Complete the square to find the roots of the quadratic function $f(x) = 5x^2 - 7x + 1$. (3 marks)

Leaving 5 in

$$f(x) = 5\left(x^2 - \frac{7}{5}x + \frac{1}{5}\right)$$

Also:

Could use formula from formula sheet.

$$0 = 5\left[\left(x - \frac{7}{10}\right)^2 - \frac{49}{100} + \frac{20}{100}\right]$$

$$0 = 5\left[\left(x - \frac{7}{10}\right)^2 - \frac{29}{100}\right]$$

$$0 = 5\left(x - \frac{7}{10}\right)^2 - \frac{29}{20}$$

$$0 = \left(x - \frac{7}{10}\right)^2 - \frac{29}{100}$$

Must solve

$$f(x) = 0$$

to find roots.

$$f(x) = 0$$

$$0 = x^2 - \frac{7}{5}x + \frac{1}{5} \quad \checkmark$$

$$0 = \left(x - \frac{7}{10}\right)^2 - \frac{49}{100} + \frac{1}{5}$$

$$0 = \left(x - \frac{7}{10}\right)^2 - \frac{29}{100} \quad \checkmark$$

$$\frac{29}{100} = \left(x - \frac{7}{10}\right)^2$$

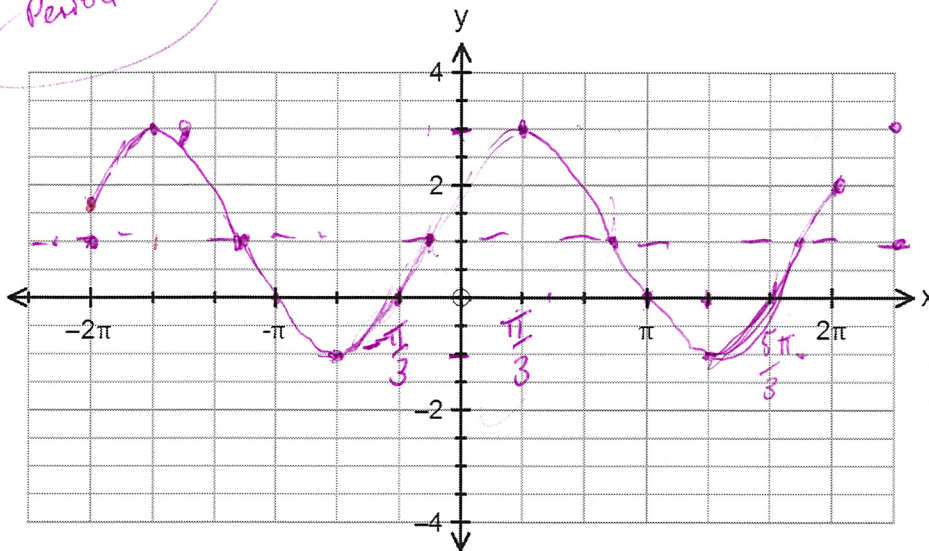
$$\frac{\pm\sqrt{29}}{10} = x - \frac{7}{10}$$

$$x = \frac{\pm\sqrt{29} + 7}{10} \quad \checkmark$$

Question 6 (7 marks)

- (a) Sketch the curve $y = 1 + 2\cos\left(x - \frac{\pi}{3}\right)$ for $x \in [-2\pi, 2\pi]$ (3 marks)

No change Period 2π



✓ Centre $y=1$

✓ max 3
min -1

Domain ✓

- (b) Hence, or otherwise, solve the following equations where $-2\pi \leq x \leq 2\pi$.

(i) $2\cos\left(x - \frac{\pi}{3}\right) = -1$

4 soln

4π
2 cycles

(2 marks)

$2\cos\left(x - \frac{\pi}{3}\right) + 1 = 0$

$0 = 1 + 2\cos\left(x - \frac{\pi}{3}\right)$

↑

(x-intercepts)

$x = -\pi, -\frac{\pi}{3}, \pi, \frac{5\pi}{3}$

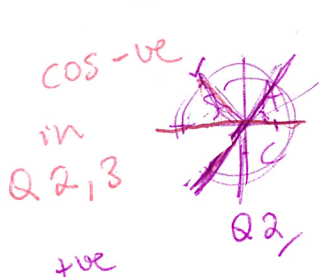
↑
Add $2\pi - \frac{\pi}{3}$

1/2 each

Recall roots from graph

(ii) $\cos x = -\frac{1}{2}$

(2 marks)



$\cos 60^\circ = \frac{1}{2}$

$\cos \frac{\pi}{3} = \frac{1}{2}$

Quad 2,3

4 soln's

$\pi - \frac{\pi}{3} = \frac{2\pi}{3}$

$\pi + \frac{\pi}{3} = \frac{4\pi}{3}$

$-\pi + \frac{\pi}{3} = -\frac{2\pi}{3}$

$-\frac{4\pi}{3}$

$\therefore x = \frac{2\pi}{3}, \frac{4\pi}{3}, -\frac{2\pi}{3}, -\frac{4\pi}{3}$

1/2 each

See next page

Hint:
You can subtract $\frac{\pi}{3}$ from answers in part (b) to get the solutions

Question 7 (11 marks)

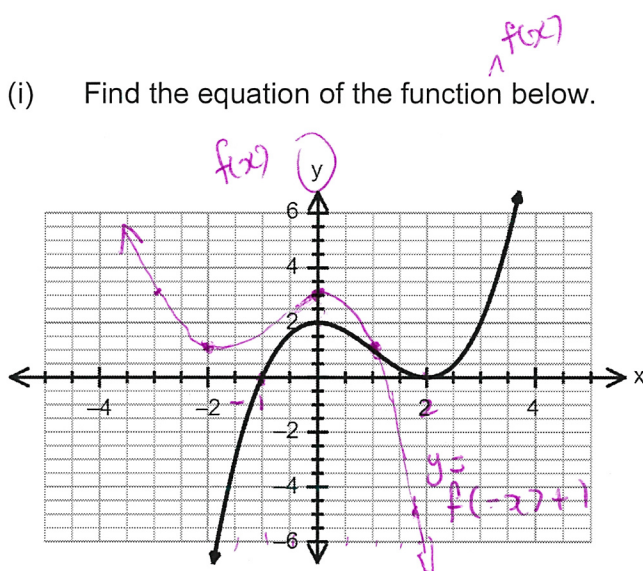
- (a) The polynomial $p(x) = x^3 + 2x^2 - 5x - 6$ has a factor $(x + 1)$.
Express $p(x)$ as the product of three linear factors.

(3 marks)

$$\begin{aligned} \therefore x &= -1 \quad p(x) = 0 \\ p(2) &= 8 + 8 - 5(2) - 6 \\ &= 0 \\ p(x) &= (x+1)(x-2)(x+3) \\ p(-3) &= -27 + 18 + 15 - 6 \\ &= 0 \quad \therefore (x+3) \text{ is a factor} \end{aligned}$$

- (b) (i) Find the equation of the function below.

(2 marks)



$$\begin{aligned} y &= a(x+1)(x-2)^2 \\ 2 &= a(1)(-2)^2 \\ 2 &= 4a \\ a &= \frac{1}{2} \quad \therefore y = \frac{1}{2}(x+1)(x-2)^2 \end{aligned}$$

- (ii) Describe the sequence of transformations that maps the graph of $y = f(x)$ onto the graph of $y = \frac{1}{2}f(2x-1)$.

(3 marks)

$$\begin{aligned} y &= \frac{1}{2}f(x) \quad \text{Vertical dilation S.F. } \frac{1}{2} \\ y &= \frac{1}{2}f(x-1) \quad \text{Horiz. translation } +1 \text{ unit (right)} \\ y &= \frac{1}{2}f(2x-1) \quad \text{Horiz. dilation S.F. } \frac{1}{2} \end{aligned}$$

- (ii) The local maximum of the function $y = f(x)$ has the coordinates $(0, 2)$.

Find the coordinates of the local maximum of the function $y = \frac{1}{2}f(2x-1)$. (1 mark)

$$\left(\frac{1}{2}, 1\right)$$

- (iii) On the same set of axes, draw the function $y = f(-x) + 1$

(2 marks)

See graph ✓

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Question 8 (6 marks)

The quadratic function $h(t) = -5t^2 + 180$ for $t \geq 0$ describes the height, h metres, above ground level of a falling tennis ball after t seconds

a) Draw a graph of $h(t)$ showing intercepts.

(2 marks)



b) How high was the ball after 1 second?

(2 marks)

$$h(1) = -5(1)^2 + 180 \\ = 175 \text{ m}$$

c) For how long was the ball above a height of 100m?

(2 marks)

$$100 = -5t^2 + 180 \\ 5t^2 = 80 \\ t^2 = 16 \\ t = \pm 4 \quad \therefore 4 \text{ sec.}$$

Question 9 (3 marks)

By using the appropriate addition formula find the exact value of $\cos\left(\frac{5\pi}{12}\right)$.

(3 marks)

$$\frac{\pi}{4} = \frac{3\pi}{12}$$

$$\cos\left(\frac{\pi}{4} + \frac{\pi}{6}\right)$$

$$A = \frac{\pi}{4}$$

$$B = \frac{\pi}{6}$$

$$\frac{\pi}{6} = \frac{2\pi}{12}$$

$$= \cos\frac{\pi}{4} \cdot \cos\frac{\pi}{6} - \sin\frac{\pi}{4} \cdot \sin\frac{\pi}{6}$$

using:

$$\cos(A+B) = \cos A \cos B \\ - \sin A \sin B$$

$$\frac{\sqrt{2}}{2} \cdot \frac{\sqrt{3}}{2} - \frac{\sqrt{2}}{2} \cdot \frac{1}{2}$$

$$= \frac{\sqrt{6}}{4} - \frac{\sqrt{2}}{4}$$

$$= \frac{\sqrt{6} - \sqrt{2}}{4} \quad \frac{\sqrt{2}(\sqrt{3} - 1)}{4}$$

End of Questions



Kennedy Baptist College
Semester One Examination 2018
Question/Answer Booklet

MATHEMATICS
METHODS UNIT 1

Section Two:
Calculator-assumed

Student Name: _____

Teacher's Name: _____

SOLUTIONS

Time allowed for this section

Reading time before commencing work:

ten minutes

Working time for paper:

one hundred minutes

Material required/recommended for this section

To be provided by the supervisor

This Question/Answer booklet

Formula Sheet (retained from Section One)

To be provided by the candidate

Standard items: pens(blue/black preferred), pencils(including coloured), sharpener, correction tape/fluid, erasers, ruler, highlighters

Special Items: drawing instruments, templates, notes on two unfolded sheets of A4 paper, and up to three calculators approved for use in the WACE examinations.

Important note to candidates

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Section Two: Calculator-assumed

65% (98 marks)

This section has **thirteen (13)** questions. Attempt **all** questions. Write your answers in the spaces provided.

Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.

- Planning: If you use the spare pages for planning, indicate this clearly at the top of the page.
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Working time: 100 minutes

Question 10 (7 marks)

The current A (amperes), varies inversely to the resistance R (ohms) in an electric circuit. When the resistance is 12 ohms, the current is 0.5 amperes.

- (a) State the constant of proportionality. $A = \frac{k}{R}$ (1 mark)

$$RA = k \quad 12 \times 0.5 = 6 \quad k = 6 \checkmark$$

- (b) Calculate the resistance if the current is 5 amperes. (1 mark)

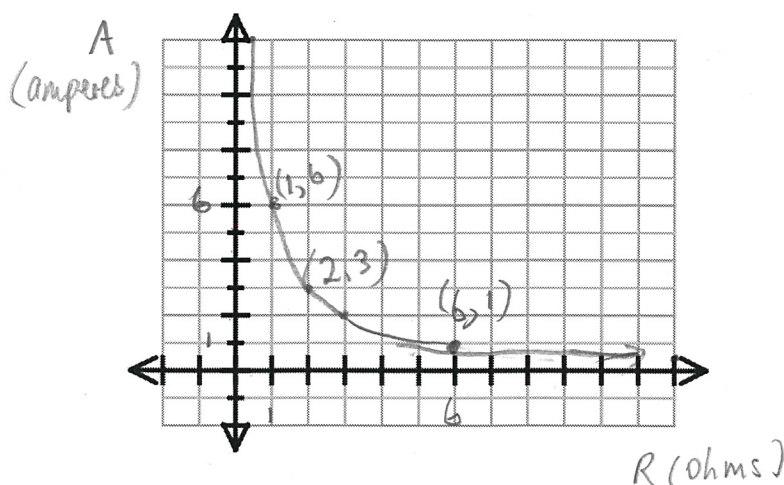
$$R = \frac{k}{A} \quad R = \frac{6}{5} = 1.2 \text{ ohms } \checkmark$$

- (c) Determine the effect on R if A is increased by 35%. (2 marks)

$$R = \frac{6}{A} \Rightarrow R = \frac{6}{1.35A} \checkmark$$

$HA=1$
 $R = 6 \Rightarrow \frac{20}{27}$
 $= 0.74 \therefore R \text{ is } 74\% \text{ of original or reduced by } 26\% \checkmark$

- (d) Draw the graph of this relationship. Label the axes. (3 marks)



label axes $\checkmark$
 shape, points $\checkmark$

Question 11 (12 marks)

The depth, x metres of a tidal river at a particular point on the river, t hours after midnight on Sunday, is modelled by $x(t) = 2\cos\frac{\pi t}{6} + 3, t \geq 0$. (Assume that midnight on Sunday corresponds to the end of Sunday).

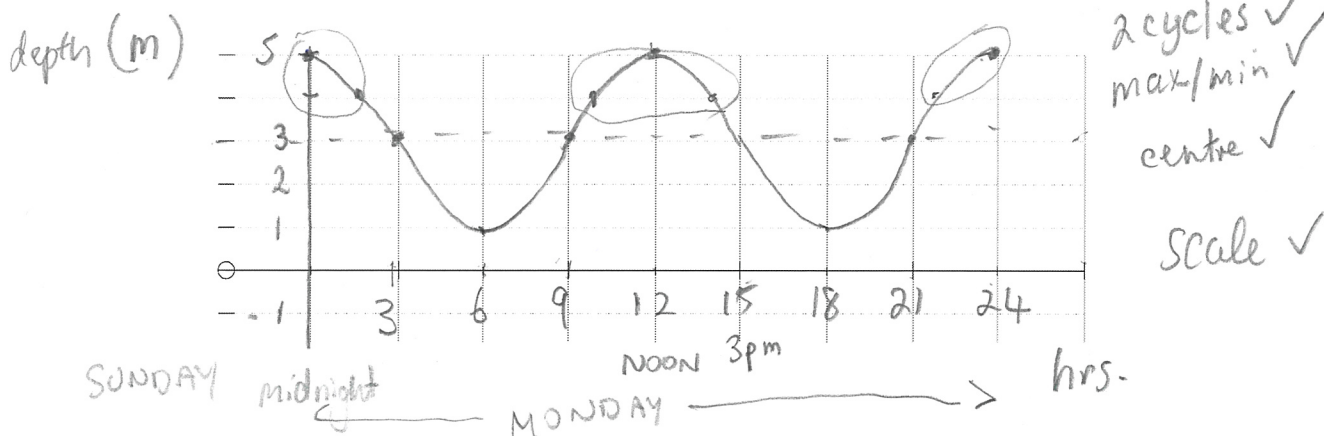
- a) Write down the minimum and maximum depth of the river for the above model. (2 marks)

max is 5m, min, 1m

- b) What is the period of the graph in hours? (2 marks)

$$P = \frac{2\pi}{\frac{\pi}{6}} = \frac{2\pi \times 6}{\pi} = 12 \text{ hours}$$

- c) Draw a graph showing the depth of the river over a 24 hour period. Show clearly labelled axes and scales for both axes. (4 marks)



- d) Find the depth, in metres, of the river at 5pm on Monday:

- i) exactly

$$t = 17$$

$$(-\sqrt{3} + 3) \text{ m}$$

- ii) correct to 1 decimal place.

$$1.3 \text{ m}$$

- e) A ship needs a depth of at least 4m to sail down the river. Find the times on Monday, when the ship can sail past this point. (2 marks)

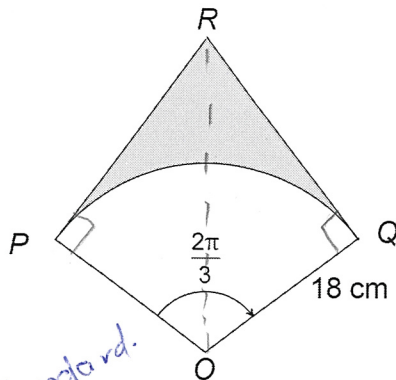
[Intervals needed]

$t = 0 \rightarrow t = 2$	} ✓	midnight to 2am	} ✓
$t = 10 \rightarrow t = 14$		10am to 2pm (1400)	
$t = 22 \rightarrow 24$		10pm to midnight (2200 → 0000)	

Question 12 (7 marks)

A sector OPQ of a circle with centre O is drawn below. The radius of the circle is 18 cm and angle POQ is $\frac{2\pi}{3}$ radians. The tangents to the circle at the points P and Q meet at point R .

$$\angle RPO = \angle RQO = \frac{\pi}{2}$$



CAS To Standard.

- (a) Determine the exact length of arc PQ .

(2 marks)

$$\begin{aligned} PQ &= r\theta \\ &= 18 \times \frac{2\pi}{3} = 12\pi \text{ cm} \end{aligned}$$

- (b) Find the exact value in radians of $\angle PRQ$.

(1 mark)

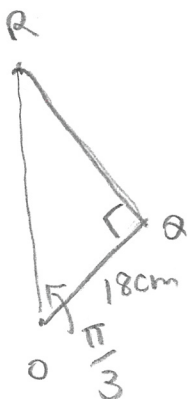
$$2\pi - \pi - \frac{2\pi}{3} = \frac{\pi}{3}$$

- (c) Find the exact area of the shaded region.

(4 marks)

Labels!

$$\begin{aligned} \text{Area (sector)} &= \frac{1}{2} r^2 \theta \\ &= \frac{1}{2} (18^2) \left(\frac{2\pi}{3} \right) \\ &= 108\pi \text{ cm}^2 \end{aligned}$$



$$\tan \frac{\pi}{3} = \frac{QR}{18}$$

$$QR = 18\sqrt{3}$$

$$\begin{aligned} \text{Area } PRQO &= 2 \times \frac{1}{2} \times 18 \times 18\sqrt{3} \\ &= 324\sqrt{3} \end{aligned}$$

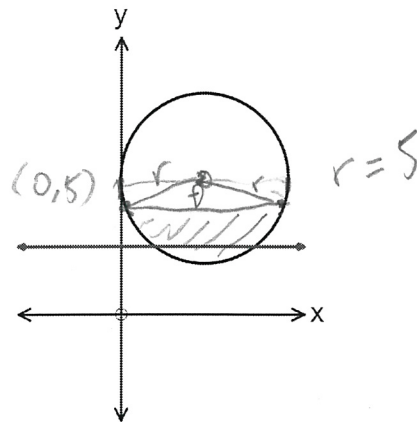
$$\begin{aligned} \therefore \text{Shaded area} \\ &= 324\sqrt{3} - 108\pi \text{ cm}^2 \end{aligned}$$

See Next Page

*None exact.
No marks if using segment / method*

Question 13 (7 marks)

The circle with centre A(5, 8) touches the y – axis as shown below.



- (a) Show clearly that the equation of the circle is $x^2 + y^2 - 10x - 16y + 64 = 0$. (2 marks)

$$\begin{aligned}(x-5)^2 + (y-8)^2 &= 25 \quad \checkmark \\ x^2 - 10x + 25 + y^2 - 16y + 64 &= 25 \quad \checkmark \\ x^2 + y^2 - 10x - 16y + 64 &= 0\end{aligned}$$

- (b) The line $y = 4$ intersects the circle at point M and N.

- (i) Determine the length of the chord MN. (2 marks)

NOT an arc.

Sub. $y = 4$ Solve using CAS.

$$x = 2, x = 8 \quad \checkmark$$

$$\therefore MN = 6 \text{ units} \quad \checkmark$$

- (ii) Find the area of the minor segment formed between MN and the circle. (3 marks)

$$A = \frac{1}{2} r^2 (\theta - \sin \theta)$$

Find θ $6^2 = 5^2 + 5^2 - 50 \cos \theta$

$$\theta = 1.2870 \quad \checkmark$$

$$\begin{aligned}A &= \frac{1}{2} (5^2) (1.287 - \sin 1.287) \quad \checkmark \\ &= 4.0875 \text{ units}^2 \quad \checkmark\end{aligned}$$

* Must be radians

Question 14 (7 marks)

- a) Determine the smallest positive solution to the equation $3 \cos(3x) = 1$, where x is in degrees, giving your solution to three significant figures. (1 marks)

$$\cos(3x) = \frac{1}{3}$$

$$3x = 70.528^\circ$$

$$x = 23.5^\circ \quad \checkmark \text{ 3 sig. fig.}$$

- b) By using the expansion of $\cos(A+B)$ show that $\cos(2x) = 2\cos^2 x - 1$ (2 marks)

$$\cos(A+B) = \cos(x+x) = \cos(2x) \quad x=A \quad x=B$$

$$= \cos x \cos x - \sin x \sin x \quad \checkmark$$

$$= \cos^2 x - \sin^2 x$$

$$= \cos^2 x - (1 - \cos^2 x) \quad \checkmark$$

$$= 2\cos^2 x - 1$$

Using:

$$\sin^2 x + \cos^2 x = 1$$

$$\leftarrow \sin^2 x = 1 - \cos^2 x \quad (\text{Sub.})$$

- c) Solve

- i) $\cos 2x = 0.7$ for $0 \leq x \leq 360$ correct to 1 decimal place. (2 marks)

Using
CAS

$$x = 157.2^\circ, x = 202.8^\circ, 337.2^\circ, 22.8^\circ$$

$\frac{1}{2}$ each

- ii) $\sin\left(\frac{x}{2}\right) = \cos\left(\frac{x}{2}\right)$ for $0 \leq x \leq 4\pi$. Give your answer in exact form in radians. (2 marks)

$$2x = \frac{\pi}{4}$$

OR USE CAS

$$\text{Solve } \left(\sin\left(\frac{x}{2}\right) = \cos\left(\frac{x}{2}\right) \mid 0 \leq x \leq 4\pi\right)$$



$0 \rightarrow 4\pi$

2 rotations

Quadrants 1 & 3

$$\text{Period} = 2\pi \div \frac{1}{2}$$

$$= 4\pi$$

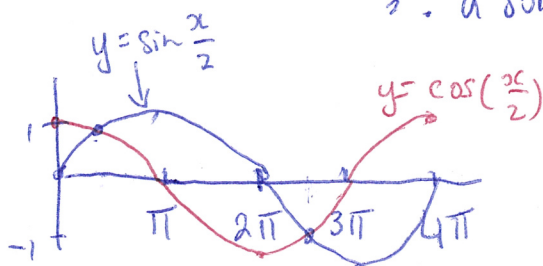
$\therefore$ 2 solutions

$$\frac{x}{2} = \frac{\pi}{4}$$

$$x = \frac{\pi}{2}$$

$$\frac{\pi}{2} + 2\pi = \frac{5\pi}{2}$$

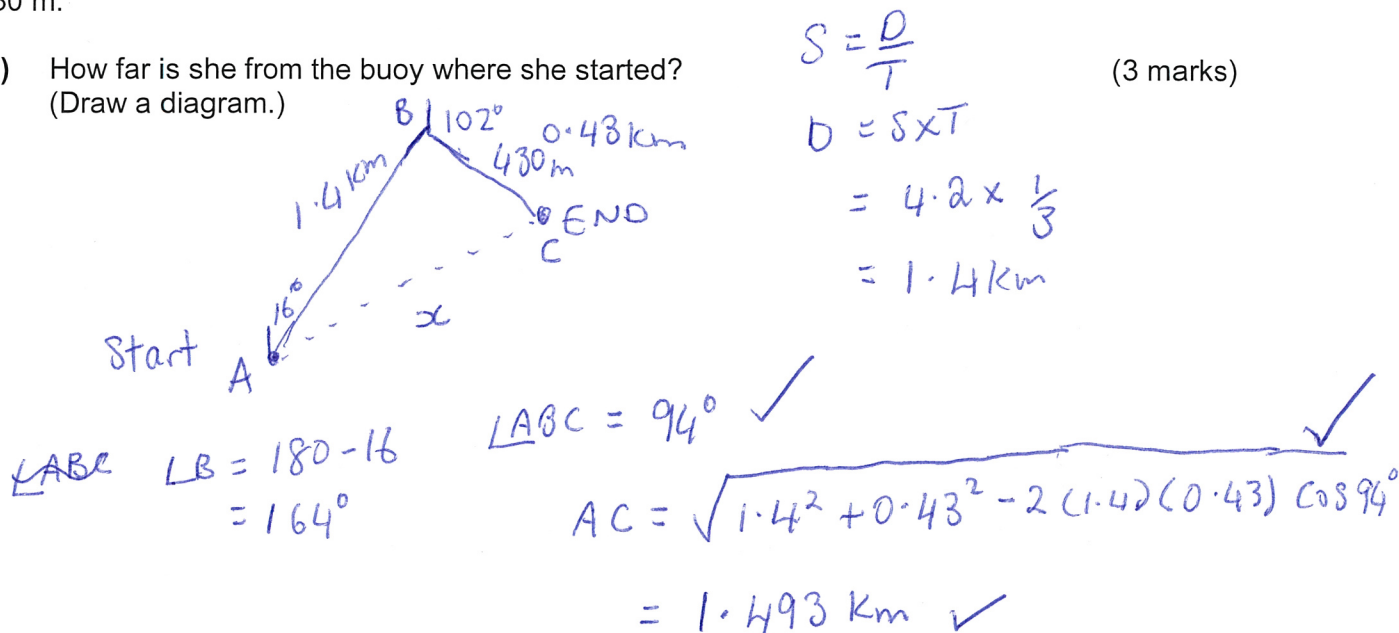
Graph $\Rightarrow$



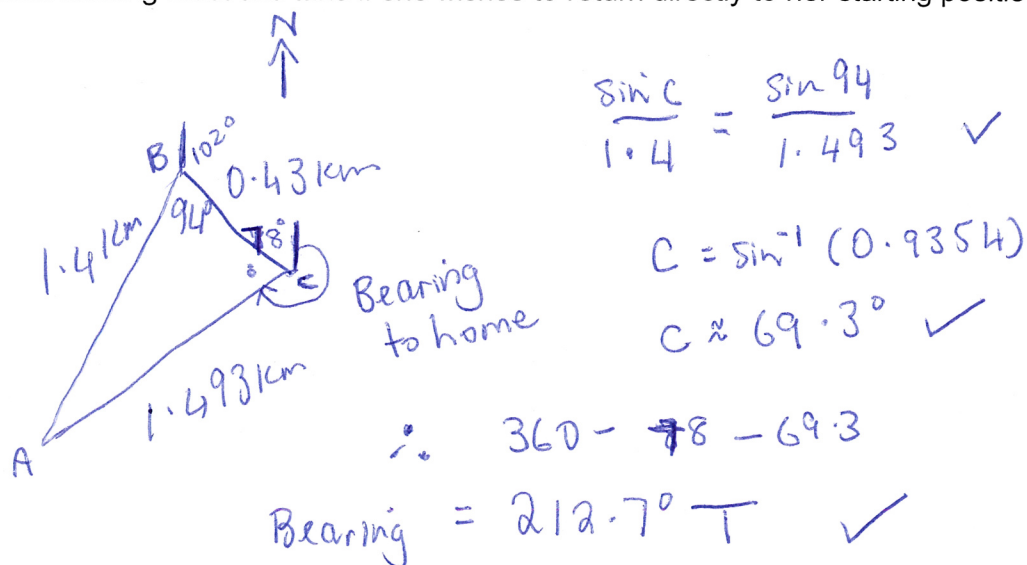
Question 15 (6 marks)

Melissa is on her stand-up board and paddles from a buoy on a bearing of 016°T for 20 minutes at a speed of 4.2 km/hr. She then changes course and paddles on a bearing of 102°T for a further 430 m.

- (a) How far is she from the buoy where she started?
(Draw a diagram.)



- (b) What bearing must she take if she wishes to return directly to her starting position? (3 marks)



Question 16 (9 marks)

The point $P(1,k)$ lies on the curve with equation $f(x) = (x^2 - 6x)(x - 2) + 3x$

- (i) Find the value of k . $f(1) = (1^2 - 6(1))(1 - 2) + 3(1)$ (1 mark)

$$\therefore k = 8 \quad \checkmark$$

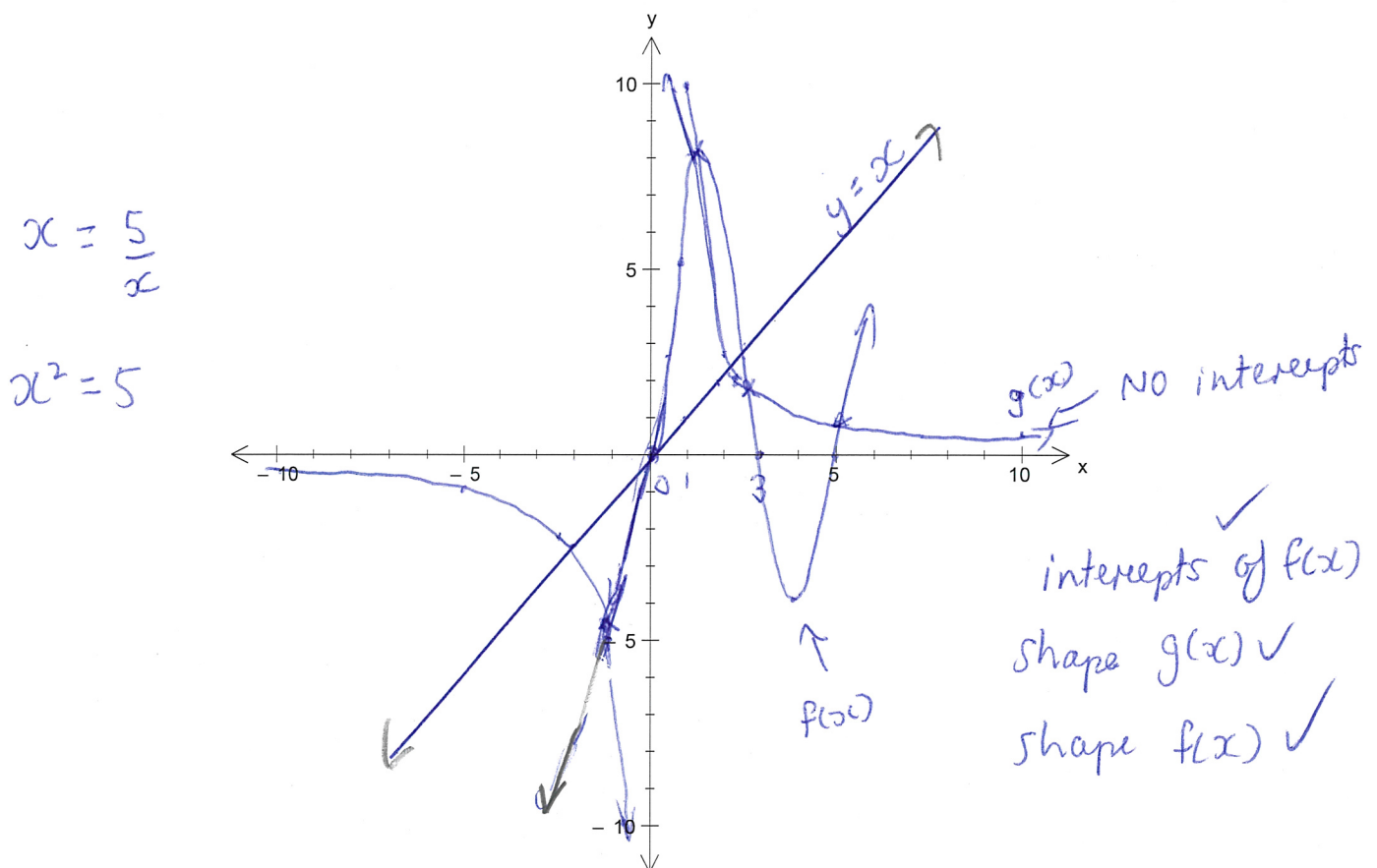
- (ii) Express $f(x)$ in a completely factorised form. (2 marks)

$$f(x) = x(x-3)(x-5) \quad \checkmark \checkmark$$

- (iii) On the axes below sketch the curves with the following equations

$$f(x) = (x^2 - 6x)(x - 2) + 3x \text{ and } g(x) = \frac{5}{x}, \text{ showing clearly where they meet the axes.}$$

(3 marks)



- (iv) Read from the graph the **number** of solutions to the equation. (1marks)

$$(x-5)(x-3) = \frac{5}{x^2}$$

4 solutions

- (v) By adding **one** suitable function on the axes, show how you solve $x^2 = 5$ and estimate the solution(s) from your graph, correct to 1 decimal place. (2.marks)

* Also parabola

$$x = \frac{5}{x}$$

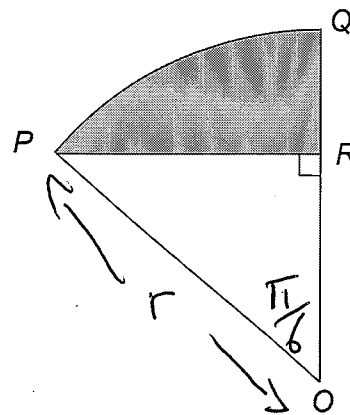
$$x^2 = 5$$

$$x \approx -2.2, x = 2.2$$

Graph of $y = x$ above

Question 17 (6 marks)

The diagram below has an arc, PQ , of a circle with centre O and radius r .
 PR is perpendicular to OQ . Angle $POQ = \frac{\pi}{6}$ radians.



- (a) Show that the area of triangle $POR = \frac{r^2\sqrt{3}}{8}$ in terms of r . (3 marks)

$$\begin{aligned} \sin \frac{\pi}{6} &= \frac{PR}{r} & \cos \frac{\pi}{6} &= \frac{OR}{r} \\ PR &= r \sin \frac{\pi}{6} & OR &= r \cos \frac{\pi}{6} \\ &= \frac{r}{2} & &= \frac{\sqrt{3}}{2} r \end{aligned}$$

$$\text{Area } \triangle POR = \frac{1}{2} \times \frac{r}{2} \times \frac{\sqrt{3}}{2} r = \frac{r^2\sqrt{3}}{8}$$

- (b) If the shaded area is $\frac{2\pi - 3\sqrt{3}}{6} \text{ cm}^2$, calculate the value of r . (3 marks)

$$\text{Shaded area} = \text{Area}(\text{sector } OPQ) - A(\triangle POR)$$

$$\frac{2\pi - 3\sqrt{3}}{6} = \frac{1}{2} r^2 \left(\frac{\pi}{6} \right) - \frac{r^2\sqrt{3}}{8}$$

Solve
by
CAS

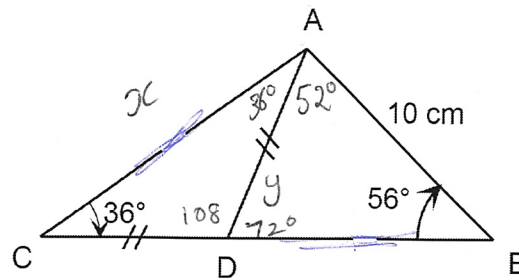
$$r = \pm 2 \text{ reject } -2$$

$$\therefore r = 2 \text{ cm}$$

Question 18 (8 marks)

The diagram shown is not drawn to scale. The lengths of AD and CD are the same. The length of AB is 10 cm.

* Edit of question needed.



- (a) If the lengths $AC = x$ and $AD = y$, find x and y correct to 1 decimal place. (4 marks)

$$\frac{y}{\sin 56^\circ} = \frac{10}{\sin 72^\circ} \checkmark$$

$$y = 8.7 \text{ cm} \checkmark$$

$$\frac{x}{\sin 108^\circ} = \frac{8.7}{\sin 36^\circ} \checkmark$$

$$x = 14.1 \text{ cm} \checkmark$$

- (b) Calculate the area of $\triangle ABC$. (2 marks)

$$A = 0.5(14.1)(10) \sin 88^\circ \checkmark$$

$$= 70.46 \text{ cm}^2 \checkmark$$

- c) A line segment AE is drawn such that AE is perpendicular to BC. What is the length of AE?

(2 marks)

$$\sin 56^\circ = \frac{AE}{10} \checkmark$$

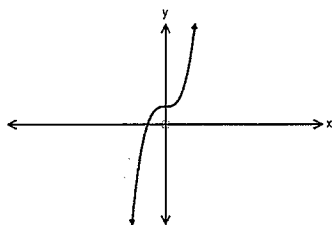
$$AE = 8.29 \text{ cm} \checkmark$$

18

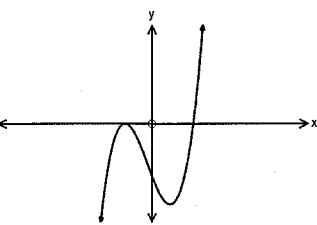
Question 19 (8 marks)

The graphs of eight functions are shown below.

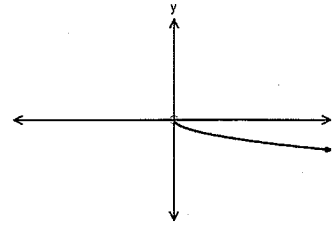
A.



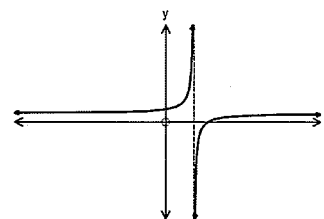
B.



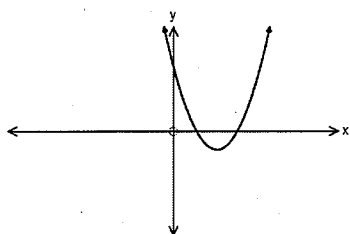
C.



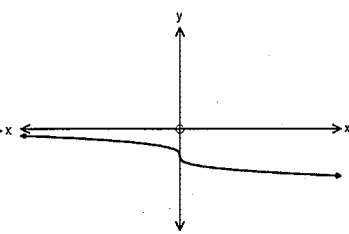
D.



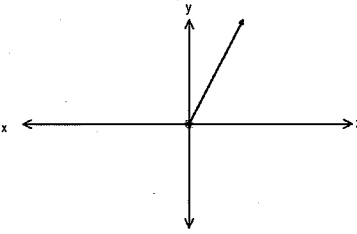
E.



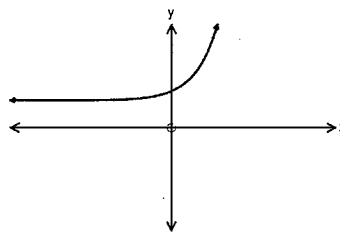
F.



G.



H.



Identify the graph(s) that:

(a) has/have horizontal asymptotes.

D, H

(1 mark)

$-\frac{1}{2}$
per
error

(b) has/have only positive domains.

C, G

(1 mark)

(c) has/have a positive constant gradient.

G

(1 mark)

(d) represent(s) direct variance.

G

(1 mark)

(e) is/are in the form $y = ak^x + b$.

H

(1 mark)

(f) pass(es) through quadrants I, II and III only.

A

(1 mark)

(g) has an axis of symmetry $x = 2$.

E

(1 mark)

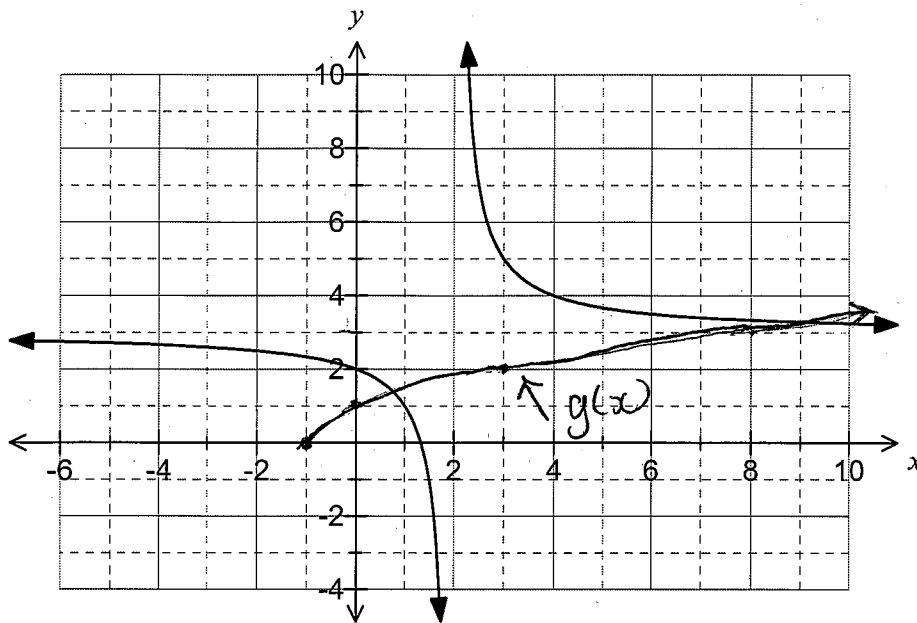
(h) has/have a degree of 3.

A, B

(1 mark)

Question 20 (6 marks)

The graph of the function $f(x) = \frac{2}{x-2} + 3$ is shown below.



(a) State the equation(s) of all asymptote(s) of $f(x)$.

(2 marks)

$$x = 2 \quad \checkmark$$

$$y = 3 \quad \checkmark$$

(b) Add the graph of $g(x) = \sqrt{x+1}$ on the axes above.

Root $\checkmark$
Y-int $\checkmark$

(2 marks)

(c) State the domain of $g(x)$.

(1 mark)

$$x \geq -1 \quad \checkmark$$

(d) State the smallest positive solution to the equation $f(x) = g(x)$.

(1 mark)

$$x = 0.795 \quad \checkmark$$

$$\text{or } 0.8$$

☺

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Question 21 (9 marks)

- (a) (i) The radius of a sphere is 15 cm. Determine the exact volume of the sphere. (1 mark)

$$V = \frac{4}{3} \pi (15)^3 = 4500 \pi \text{ cm}^3 \quad \checkmark$$

- (ii) The volume of a sphere is directly proportional to the cube of its radius. Using the information above in (i) show that the value of the constant of proportionality is $\frac{4\pi}{3}$. (1 mark)

$$V = K r^3$$

$$\therefore K = \frac{4\pi}{3} \quad \checkmark$$

- (iii) Determine the angle of inclination between the graph of the direct variance function $V = kr^3$ and the positive r^3 -axis. (2 marks)

$$\tan \theta = \frac{4\pi}{3} \quad \checkmark$$

$$\theta = 76.57^\circ \quad \checkmark \quad \text{radians or degrees}$$

- (b) The area of a room is 20m^2 . If the length is increased by 3m and the width is increased by 1m , the room will double in area. Formulate a quadratic equation, and hence determine the original dimensions of the room. (3 marks)

$$\begin{array}{|c|} \hline y \\ \hline \end{array} \times \quad xy = 20 \Rightarrow y = \frac{20}{x} \quad (1)$$

$$(x+1) \begin{array}{|c|} \hline y+3 \\ \hline \end{array} = 40 \quad (2) \quad \checkmark$$

$$(1) \rightarrow (2)$$

$$(x+1) \left(\frac{20}{x} + 3 \right) = 40 \quad \checkmark$$

$$3x^2 - 17x + 20 = 0 \quad x = \frac{5}{3} \text{ or } x = 4 \quad \checkmark$$

$$\therefore 4\text{m} \times 5\text{m} \quad \checkmark$$

$$\frac{5}{3} \times 12\text{m} \quad \checkmark$$

- (c) State the domain and range of the function $y = \frac{x+1}{x}$. (2 marks)

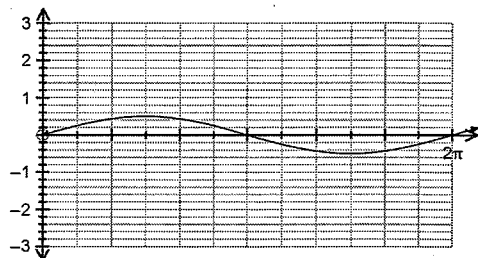
$$\text{Domain } \{x: x \in \mathbb{R}, x \neq 0\} \quad \checkmark$$

$$\text{Range: } \{y: y \in \mathbb{R}, y \neq 1\} \quad \checkmark$$

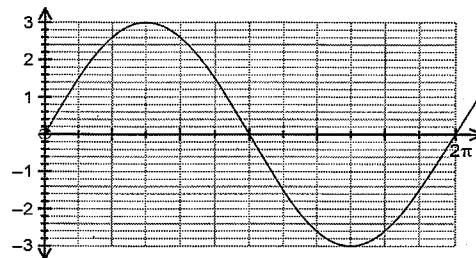
Question 22 (6 marks)

Sound waves can be modelled on sinusoid functions. The following graphs were drawn after experiments with sound waves were conducted.

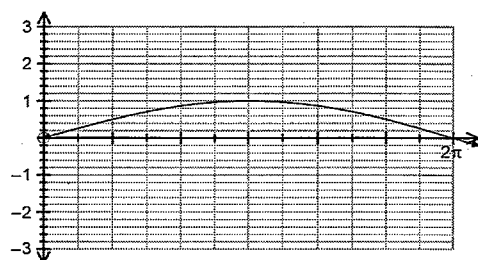
A: Quieter



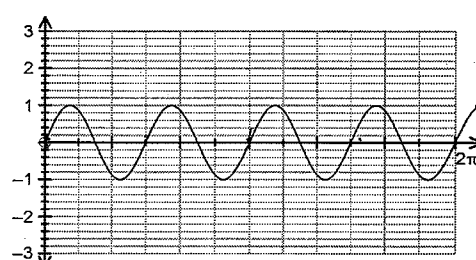
B: Louder



C: Deeper pitch



D: Higher pitch



State the equation of each graph and discuss the changes in amplitude and period to achieve the different sounds.

✓ A : $y = \frac{1}{2} \sin x$
 ✓ B : $y = 3 \sin x$

Amplitude change of $\frac{1}{2}$ to 3 $\Rightarrow \therefore$ Increase
 = Louder

✓ C : $y = \sin \frac{x}{2}$
 ✓ D : $y = \sin 4x$

Period of π to $\frac{\pi}{2}$

Higher pitch $\Rightarrow$ Smaller period

(ie frequency is higher)

End of Questions

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